

Diophantine quintuple conjecture

OPG-16555

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Definition A set of m positive integers $\{a_1, a_2, \dots, a_m\}$ is called a Diophantine m -tuple if $a_i \cdot a_j + 1$ is a perfect square for all $1 \leq i < j \leq m$.

Conjecture (1) Diophantine quintuple does not exist.

It would follow from the following stronger conjecture [Da]:

Conjecture (2) If $\{a, b, c, d\}$ is a Diophantine quadruple and $d > \max\{a, b, c\}$, then $d = a + b + c + 2bc + 2\sqrt{(ab + 1)(ac + 1)(bc + 1)}$.

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